

Question	Answers	Additional comments/Guidelines	Mark	AO
11.1	Idea that it stops the microphone picking up noises in the background/other sounds ✓		1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
11.2	191 (μs) ✓	CAO	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.3	($\pm$) 15 (μs) ✓	CAO	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.4	$3.9(3) \times 10^3 \text{ (m s}^{-1}\text{)}$ ✓	Accept 2 or 3 sf only	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.5	One of the percentage uncertainties correctly calculated ✓ 9.2(%) ✓	Allow decimal or percentage Expect to see 1.3% and 7.9% accept 1 or 2 sf only	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
11.6	The amplitude of the sound pulse gets weaker/smaller (as it travels) so would not trigger microphone B ✓ Microphone sensitivity needs to be high so this might trigger it with any noise ✓	Accept pulse/sound volume too low to measure If no other mark awarded, accept for one mark idea that uncertainty in t would increase	2	2 × AO4
Total			8	

Question	Key	Answer	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	0.03 (V) ✓	condone 2 sf answer	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	steepest line and shallowest line drawn ✓	1.67 and 1.60 on RHS of Fig. 14 4.45 and 4.65 on bottom of Fig. 14 Min line through top of 1st plot and bottom of last Max line through bottom of 3rd plot and top of last	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	evidence of using shallowest line ✓ uses $\Delta f \geq 1.0 \times 10^{14}$ to determine gradient OR correct read offs for a pair of data points ✓ method to determine h: uses $h = e \times \text{gradient}$ OR use of simultaneous equations ✓ answer that rounds to 4.7×10^{-34} (J s) ✓	If only 1 line is drawn on Fig. 14, MP2 and MP3 can still be accessed. Expect gradient around 2.9×10^{-15} (N.B. Gradient of steepest line $\sim 3.6 \times 10^{-15}$) Allow for MP3: Rearranges equation $V = \frac{hf}{e} - \frac{\phi}{e}$ with comparison to $y = mx + c$ Max 2 or 3 sf. Allow answer that rounds to 4.8×10^{-34} (J s).	4	1 × AO3 1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Max 2 from: ✓ ✓ determine gradient of steepest line OR of line of best fit uses gradients of two lines to determine an absolute uncertainty use their absolute uncertainty/mean $\times 100$	Condone taking gradient to be equal to h. MP2: e.g. $\frac{1}{2} [m(\text{steepest}) - m(\text{shallowest})]$	2	AO4
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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16	A	2.7%	AO4
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17	B	1.4%	AO4
		E_v 	

Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	0.85 (s) ✓	2 sf only	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Determines half of the range in Table 1 = 0.095 s ✓ 11 ✓	MP1: condone using full range divided by mean MP2: Accept 11%. Accept 10 (or 10%) from a correct method. 1 or 2 sf only.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	$\frac{1.63}{0.83}$ to get 1.96 m s ⁻¹ ✓ Uses their percentage uncertainty with their speed ✓	MP2: 1 or 2 sf only. Expect 0.2 m s ⁻¹ . Allow ecf from 10.2	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Idea that parallax error will result in a larger measured time _{1a}✓ Larger measured time leads to: smaller calculated (mean) terminal speed (for same height/1.63 m) or smaller percentage uncertainty in (the incorrect) time _{2a}✓ OR Idea that terminal speed is not be reached by 1.63 m _{1b}✓ (so) overall terminal speed will be lower or measured time would be larger _{2b}✓	MP1: Accept a description of parallax error e.g. "line is below student A's eyes" Accept idea that results are unaffected if student A crouches to remove parallax error. Reject idea that measurement of height is affected. Accept comments about redrawing the line on the wall to remove parallax error.	2	AO4
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Micrometer (screw gauge) / digital Vernier caliper ✓		1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Zero before use OR Take readings and subtract reading when micrometer fully closed ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	To reduce random errors / to identify and eliminate anomalies OR Idea of determining whether wire has circular cross-section / uniform thickness ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Any two from: ✓✓ • calculates mean d (expect 0.17 mm) • uses $\frac{1}{2} \frac{\text{range}}{\text{their } \bar{d}} (\times 100)$ to find % uncertainty in d • $2 \times$ their % uncertainty in d to find uncertainty in A 12% ✓	Only first bullet point and final answer require anomaly to be removed	3	1 × AO2 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	% uncertainty in length 1.2% ✓ Adds their answer to their value in 10.4 ✓		2	AO3

Total			8	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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